

Cosmology: Extra Problems

A. Free Field in De Sitter

Consider a massive scalar field in D -dimensional de Sitter space:

$$S = \frac{1}{2} \int d^D x a^{D-2}(\eta) \left[(\phi')^2 - (\nabla \phi)^2 - m^2 a^2 \phi^2 \right], \quad (1)$$

where $a(\eta) = -1/(H\eta)$.

1. Show that the action of the canonically normalized field $v \equiv a^{(D-2)/2} \phi$ is

$$S = \frac{1}{2} \int d^D x \left[(v')^2 - (\nabla v)^2 - \frac{1}{\eta^2} \left(\frac{m^2}{H^2} - \frac{D(D-2)}{4} \right) v^2 \right]. \quad (2)$$

This is the action of a free field in Minkowski space with a time-dependent mass.

2. We first consider the special case $m = \frac{1}{2} \sqrt{D(D-2)}H$, for which the mass term vanishes. Starting from the Minkowski mode function

$$v_k^{\text{Mink}}(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta}, \quad (3)$$

derive the mode function $f_k(\eta)$ of the field ϕ .

3. Next, we specialize to a massless field in $D = 4$ dimensions.

- (a) Show that at sufficiently early times the de Sitter equation of motion for $v_k(\eta)$ becomes the Minkowski equation of motion. We therefore expect $v_k(\eta)$ to behave like (3) at early times.
- (b) The most general solution for the mode function in de Sitter is

$$v_k(\eta) = \alpha \left(1 - \frac{i}{k\eta} \right) e^{-ik\eta} + \beta \left(1 + \frac{i}{k\eta} \right) e^{ik\eta}, \quad (4)$$

where α and β are constants of integration. Find the values of α and β , so that v_k and $\partial_\eta v_k$ match v_k^{Mink} and $\partial_\eta v_k^{\text{Mink}}$ at some reference time η_* . Show that in the limit $\eta_* \rightarrow -\infty$ these coefficients become

$$\alpha = \frac{1}{\sqrt{2k}}, \quad \beta = \frac{1}{\sqrt{2k}} \frac{e^{-2ik\eta_*}}{2(k\eta_*)^2} \rightarrow 0. \quad (0.1)$$

This is the Bunch–Davies mode function.

B. Surprising Zeroes

Consider a massive scalar field in D -dimensional de Sitter space:

$$S = \int d^D x a^{D-2}(\eta) \left(\frac{1}{2} [(\phi')^2 - (\nabla \phi)^2] - \frac{1}{2} m^2 a^2 \phi^2 - \frac{\lambda}{4!} a^2 \phi^4 \right), \quad (1)$$

where $a(\eta) = -1/(H\eta)$. We specialize to the mass $m = \frac{1}{2} \sqrt{D(D-2)}H$ (see Problem A).

1. Show the action for the canonically normalized field $v \equiv a^{(D-2)/2}\phi$ becomes the action of a massless field in flat space with a time-dependent coupling $\lambda(\eta) \propto a^{4-D}(\eta)$.
2. Write down the correctly normalized positive-frequency solution for $v_k(\eta)$. Use this to obtain the mode function $f_k(\eta)$ for the original field ϕ .
3. Next, we wish to compute the four-point function

$$\langle \phi_{\mathbf{k}_1}(\eta_*) \phi_{\mathbf{k}_2}(\eta_*) \phi_{\mathbf{k}_3}(\eta_*) \phi_{\mathbf{k}_4}(\eta_*) \rangle', \quad (2)$$

where $\eta_* \approx 0$ is the boundary of the spacetime.

- (a) Describe the Feynman rules to compute an n -point function in the in-in formalism.
- (b) Derive the four-point function (2) at leading order in λ . Do you notice a difference between even and odd D ?

Hint: The bulk-to-boundary propagator on the positive branch is

$$K_+(k, \eta) = f_k^*(\eta) f_k(\eta_*).$$

You may also use that

$$\int_{-\infty}^0 (-\eta)^{D-4} e^{iK\eta} = \frac{i^{D-3}}{K^{D-3}} \Gamma[D-3],$$

where $\Gamma[D-3]$ is the Euler gamma function.

C. Anharmonic Oscillator

Consider the anharmonic oscillator, with action

$$S[\Phi] = \int dt \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} \omega^2 \Phi^2 - \frac{1}{3} \lambda \Phi^3 \right), \quad (1)$$

where ω and λ are constant parameters. A formal solution of the equation of motion for the anharmonic oscillator is

$$\Phi_{\text{cl}}(t) = \phi e^{i\omega t} + i \int dt' G(t, t') (-\lambda \Phi_{\text{cl}}^2(t')), \quad (2)$$

where $\phi e^{i\omega t}$ is the solution to the free equation of motion and $G(t, t')$ is the Green's function.

- (a) For small λ , that solution can be written as a perturbative expansion $\Phi_{\text{cl}}(t) = \sum_n \lambda^n \Phi^{(n)}(t)$. Use the Green's function

$$G(t, t') = \frac{1}{2\omega} \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t+t')} \right) \quad (3)$$

to derive $\Phi^{(1)}(t)$.

- (b) The on-shell action is

$$S[\Phi_{\text{cl}}] = \frac{i\omega}{2} \phi^2 - \frac{\lambda}{3} \int dt \Phi_{\text{cl}}^3(t) - \frac{i\lambda^2}{2} \int dt dt' G(t, t') \Phi_{\text{cl}}^2(t') \Phi_{\text{cl}}^2(t). \quad (4)$$

Derive the on-shell action up to order λ^2 and show that the wavefunction of the oscillator (in the saddle-point approximation) is

$$\Psi[\phi] \approx e^{iS[\Phi_{\text{cl}}]} = \exp \left(-\frac{\omega}{2}\phi^2 - \frac{\lambda}{9\omega}\phi^3 + \frac{\lambda^2}{72\omega^3}\phi^4 + \dots \right). \quad (5)$$

Use this to compute $\langle \phi^3 \rangle$ and $\langle \phi^4 \rangle$.

D. Wavefunction and Correlator

Consider a massless scalar in de Sitter space with the cubic interaction $\dot{\Phi}^3$.

1. Using the in-in formalism, derive the bispectrum $\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle$, where $\phi_{\mathbf{k}} \equiv \Phi_{\mathbf{k}}(\eta = 0)$.
2. Now derive the wavefunction coefficients ψ_2 and ψ_3 . Use the Born rule to derive the relation

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle' = \frac{\text{Re } \psi_3(\mathbf{k}_a)}{\prod_{a=1}^3 \text{Re } \psi_2(k_a)}. \quad (1)$$

Verify by direct substitution that your wavefunction coefficients give the correct bispectrum derived in part 1).

E. Breaking Boosts

Consider the (de Sitter breaking) interaction

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{2} \dot{\Phi}^2 \chi, \quad (1)$$

where Φ is a massless scalar and χ is a conformally coupled scalar. Compute the four-point wavefunction coefficient ψ_4 for massless scalars coming from the exchange of χ . You should find

$$\psi_4 = -\frac{\lambda^2}{H^2} k_1^2 k_2^2 k_3^2 k_4^2 \frac{2E_L E_R + E(E_L + E_R) + E^2}{E^3 E_L^2 E_R^2}, \quad (2)$$

where $E \equiv k_{12} + k_{34}$, $E_L \equiv k_{12} + k_I$, $E_R \equiv k_{34} + k_I$. Can you relate this to the four-point function of a massless scalar in flat space?

F. Breaking Unitarity

Consider the (non-unitary) action

$$S = \int d^4x \left(-\frac{1}{2}(\partial\Phi)^2 + \frac{1}{2}(\partial\chi)^2 - \lambda\Phi^2\chi \right), \quad (4)$$

which couples together a healthy field Φ , with a ghostly field χ . Compute the four-point wavefunction coefficient for Φ arising from exchanging a χ field, and verify that it does *not* satisfy the cutting rule shown in part (c) of Problem 3.

G. Bootstrapping from Unitarity

In Problem D, you computed the three-point wavefunction coefficient for a massless scalar in de Sitter space with the cubic interaction $\dot{\Phi}^3$. You should have found that

$$\psi_3(k_1, k_2, q) = -2 \frac{(k_1 k_2 q)^2}{(k_1 + k_2 + q)^3}. \quad (1)$$

We will now use this to bootstrap the four-point wavefunction coefficient ψ_4 . Our main tool will be the unitarity cut

$$\tilde{\psi}_4 = -P(q) \left(\tilde{\psi}_3 \times \tilde{\psi}_3 \right), \quad (2)$$

where $\tilde{\psi}_n \equiv \psi_n(\mathbf{k}) + \psi_n^*(-\mathbf{k})$.

1. First, show that

$$\tilde{\psi}_{3, \dot{\Phi}^3}^{(\text{dS})} = 4q^3 (k_1 k_2)^2 \frac{(3k_{12}^2 + q^2)}{(k_{12}^2 - q^2)}. \quad (3)$$

It is useful to compare this to the three-point wavefunction coefficient for a massless scalar in *flat space* with the cubic interaction Φ^3 . Show that

$$\tilde{\psi}_{3, \Phi^3}^{(\text{dS})} = (k_1 k_2 q)^2 \partial_{k_{12}}^2 \tilde{\psi}_{3, \Phi^3}^{(\text{flat})}, \quad (4)$$

where $\psi_{3, \Phi^3}^{(\text{flat})} = (2q)/(k_{12}^2 - q^2)$.

2. Using (2), show that the cut of the four-point wavefunction coefficient is

$$\tilde{\psi}_{4, \dot{\Phi}^3}^{(\text{dS})} = (k_1 k_2 k_3 k_4 q)^2 \partial_{k_{12}}^2 \partial_{k_{34}}^2 \tilde{\psi}_{4, \Phi^3}^{(\text{flat})}. \quad (5)$$

Acting with $(k_1 k_2 k_3 k_4 q)^2 \partial_{k_{12}}^2 \partial_{k_{34}}^2$ on $\psi_{4, \Phi^3}^{(\text{flat})} = (E E_L E_R)^{-1}$ then gives $\psi_{4, \dot{\Phi}^3}^{(\text{dS})}$ up to a contact term.

3. Show that

$$\begin{aligned} \psi_4 = 2q(k_1 k_2 k_3 k_4)^2 & \left[\frac{6}{E^5} \left(\frac{1}{E_L} + \frac{1}{E_R} \right) + \frac{3}{E^4} \left(\frac{1}{E_L^2} + \frac{1}{E_R^2} \right) \right. \\ & \left. + \frac{1}{E^3} \left(\frac{1}{E_L^3} + \frac{1}{E_R^3} \right) - \frac{1}{E_L^3 E_R^3} \right]. \end{aligned} \quad (6)$$

An explicit computation would give the same result with an extra $-12/(qE^5)$ inside the square bracket.